

Two Incomparable Denominator Exclusions for $\sum_{n \geq 2} (n! - 1)^{-1}$

Finite certificates and exact rationality criteria for Erdős Problem #68

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Problem 0.1 (Erdős #68). Is

$$S = \sum_{n \geq 2} \frac{1}{n! - 1}$$

irrational?

ABSTRACT

Every rational representation $S = a/q$, $q > 0$, of $S = \sum_{n \geq 2} (n! - 1)^{-1}$ satisfies

$$q \nmid 299999!, \quad q \geq 2^{39990} > 10^{12038}.$$

The first exclusion follows from an exact strict-successor carry test; the second from an exact continued-fraction computation. The central difficulty is to retain denominator information after cancellation and then compare it with the positive tail. We exhibit actual prime-power cancellation and show that, at fixed factorial moment, integral channel corrections change the residual only by an integer. The companion constant $S - e + 2$ gives an exact factorial-orbit criterion; the remaining assertion is a [cofinal miss of the strict factorial successor](#).

1 The denominator exclusions

Let $S = \sum_{n \geq 2} (n! - 1)^{-1}$ and

$$H_m = \sum_{n=2}^m \frac{1}{n! - 1}, \quad Z_m = \lfloor m! H_m \rfloor + 1, \quad b_m = mZ_{m-1} + 1 - Z_m.$$

The divisibility exclusion in the abstract rests on the implication

$$S = a/q, \quad q \mid (m-1)!, \quad m \geq 3 \implies b_m = 1. \quad (1)$$

Its proof is the factorial-tail integrality argument that Hančl and Tijdeman use for factorial series with integer coefficients [6, Lemma 2.1 and the following remark, p. 385]; for

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

the denominators $n! - 1$ the tail estimate is proved directly. Indeed, $0 < m!(S - H_m) < 1$, by comparison with the telescoping series $\sum_{n>m} (n-1)/n! = 1/m!$. Thus $m!S$ is the unique integer strictly above $m!H_m$. The same argument at $m-1$ gives $Z_m = m!S = mZ_{m-1}$, proving (1). A non-unit carry at index m therefore forces **every rational denominator to be at least m** , since any smaller positive q would divide $(m-1)!$. The exact value $b_{300000} \neq 1$ therefore excludes every divisor of $299999!$ as a denominator of S . Equivalently, the factorial index $\kappa(q) = \min\{r \geq 1 : q \mid r!\}$ must satisfy $\kappa(q) \geq 300000$. This index, rather than the largest prime factor, is the relevant measure of factorial divisibility; compare Sondow's use of it for e [15, §3, Theorem 1]. His approximation bound for e is not being applied to S . In particular, the exclusion does not cover every 299999 -smooth denominator: sufficiently high powers of a small prime need not divide $299999!$.

The independent continued-fraction exclusion bounds the denominator's size. Neither restriction implies the other: a prime between 299999 and 599998 satisfies the divisibility restriction and fails the size restriction, whereas $299999!$ does the reverse. Both conclusions are finite; they do not decide the irrationality question posed by Erdős [1, p. 102] and listed in Bloom's catalogue [2]. Their computational certificates are specified in §8.

A first reading may follow the two finite certificates in §8, the exact equivalence in Theorem 2.1, and the obstruction in (9). The channel calculations explain why that obstruction persists; they are not premises of either finite exclusion.

Adjacent factorial differences affect exactly their divisor channels. This makes the channel equations solvable by integral triangular elimination. Solving them, however, does not make the residual small: at a fixed factorial moment every correction changes it by an integer. Sections 2–4 identify the exact carry test and this limitation of channel cancellation. Section 5 then shows, at the prime 139, why a large common denominator need not survive reduction; Section 6 states the additional real comparison needed.

2 The fixed companion orbit

The termwise identity

$$\frac{1}{n! - 1} = \frac{1}{n!} + \frac{1}{n!(n! - 1)}$$

replaces the changing prefixes by the factorial orbit of one fixed number,

$$C = \sum_{n \geq 2} \frac{1}{n!(n! - 1)}.$$

Absolute convergence gives

$$C + (e - 2) = S. \tag{2}$$

The endpoint of the exponential prefix contributes residue 1 modulo m ; its positive tail removes another unit on taking the floor. This accounts for the distinguished residue -2 in the following theorem.

Canonical factorial digits come from Cantor's factorial expansion [3], in which a real number is rational exactly when its digits vanish from some index on. Galambos treats

the rationality of Cantor series in [4, Ch. II, §2.1, pp. 21–22], and Koepe and Schmiersau prove the irrationality direction for nonterminating factorial expansions whose digits are not eventually maximal [5, Example 3.2, p. 121]. The following theorem is the analogue for the companion $C = S - e + 2$, whose distinguished digit is $m - 2$.

Theorem 2.1 (fixed companion-orbit rationality boundary). *The following statements are equivalent:*

1. $S \in \mathbb{Q}$;
2. $(\lfloor m!C \rfloor + 2) \bmod m = 0$ for every sufficiently large m .

Consequently,

$$S \notin \mathbb{Q} \iff (\forall B)(\exists m > B) (\lfloor m!C \rfloor + 2) \bmod m \neq 0. \quad (3)$$

The equivalence and its cofinal form are [companion orbit boundary](#).

Proof. First suppose $S = a/q$ with $q > 0$, and take m so large that $q \mid (m-1)!$. Write

$$E_m = m! \sum_{2 \leq n \leq m} \frac{1}{n!}, \quad \varepsilon_m = m! \sum_{n > m} \frac{1}{n!}.$$

Then E_m is an integer, $0 < \varepsilon_m < 1$, and $m!S$ is an integer divisible by m . Multiplying (2) by $m!$ therefore gives

$$\lfloor m!C \rfloor = m!S - E_m - 1.$$

Every summand of E_m except the endpoint $m!/m! = 1$ is divisible by m . Thus $E_m \equiv 1 \pmod{m}$ and $\lfloor m!C \rfloor \equiv -2 \pmod{m}$.

Conversely, assume the displayed congruence from some index onward. Let $d_m(C) = \lfloor m!C \rfloor - m\lfloor (m-1)!C \rfloor$ be the canonical factorial digit. Since $0 \leq d_m(C) < m$, for $m \geq 3$ the congruence is equivalent to

$$d_m(C) = m - 2.$$

Choose N beyond the exceptional indices. The canonical factorial expansion of C then has the form

$$C = \lfloor C \rfloor + \sum_{m=2}^N \frac{d_m(C)}{m!} + \sum_{m>N} \frac{m-2}{m!}.$$

Adding $e - 2 = \sum_{m \geq 2} 1/m!$ and using the telescoping identity

$$\sum_{m>N} \frac{m-1}{m!} = \sum_{m>N} \left(\frac{1}{(m-1)!} - \frac{1}{m!} \right) = \frac{1}{N!}$$

gives

$$S = \lfloor C \rfloor + \sum_{m=2}^N \frac{d_m(C) + 1}{m!} + \frac{1}{N!},$$

which is rational. Negating the eventual statement yields the cofinal formulation above. $\square$

Remark. The remaining assertion is cofinal escape of this actual factorial orbit from -2 . The theorem identifies the required event without supplying it.

Here and below factorial digits are the floor-defined canonical digits. The alternative representation with an eventually maximal tail is not used. Thus “eventually” means at every sufficiently large integer index, whereas “cofinally” means at arbitrarily large indices; neither is a statement about a finite experimental range.

2.1 The carry and its wrap correction

The canonical factorial digit is $d_m(C) = \lfloor m!C \rfloor - m \lfloor (m-1)!C \rfloor$. For $C_m = \sum_{n=2}^m 1/(n!(n!-1))$, put

$$\delta_m = m!(C - C_m), \quad \sigma_m = \mathbf{1}_{\{m!C < \delta_m\}}.$$

The positive tail satisfies $0 < \delta_m < 1/((m+1)! - 1)$. Subtracting it crosses an integer exactly when $\sigma_m = 1$, so for $m \geq 3$

$$b_m = m - 1 - d_m(C) + \sigma_m - m\sigma_{m-1}. \quad (4)$$

This identity is unconditional. Equality $\{m!C\} = \delta_m$ gives no wrap: the shifted prefix is then an integer. The hypothesis needed to remove the correction is $\sigma_m = \sigma_{m-1} = 0$, not merely a small positive tail. For integer-coefficient factorial series, Hančl and Tijdeman prove integrality of normalised rational tails [6, Lemma 2.1 and the following remark, p. 385], and classify the polynomial-coefficient case [6, Theorem 3.1 and Corollary 3.1, pp. 390–391]. The denominators here are $n! - 1$; (4) records the extra endpoint information required after passage to the companion.

Theorem 2.2 (exact carry characterisation). *The following conditions are equivalent:*

$$S \notin \mathbb{Q}, \quad (\forall B)(\exists m > B) b_m \neq 1, \quad (\forall B)(\exists m > B) m \nmid Z_m.$$

In particular, cofinal non-unit carries imply irrationality; equivalently, the original problem is the strict-successor criterion.

The pair of equivalences is [carry characterisation](#).

Proof. Rationality forces eventual unit carries by (1). Conversely, if $b_m = 1$ eventually, then $Z_m/m!$ is eventually constant. Since

$$H_m < Z_m/m! \leq H_m + 1/m!,$$

that rational constant is S . Finally writing $x = (m-1)!H_{m-1}$ gives $b_m = m - 1 - \lfloor m\{x\} + 1/(m! - 1) \rfloor$, hence $-1 \leq b_m \leq m - 1$. For $m \geq 3$, these bounds together with $Z_m = mZ_{m-1} + 1 - b_m$, imply $m \mid Z_m$ exactly when $b_m = 1$. $\square$

3 An integral basis for factorial channels

We seek integer combinations that cancel selected denominator contributions without cancelling their factorial moment. To keep track of these two requirements, for a finite integer vector λ define

$$W_{d,n} = \frac{n!}{(d!)^{\lfloor n/d \rfloor}}, \quad M(\lambda) = \sum_n \lambda_n n!, \quad V_d(\lambda) = \sum_n \lambda_n W_{d,n}.$$

The **weights are integers**. We first allow the auxiliary coordinate e_1 , then impose the manuscript support $n \geq 2$. This coefficient convention introduces no term $1/(1! - 1)$ into S .

All vector sums are finite unless explicitly called residual series. The integer $d \geq 2$ labels a channel and n labels a support index; M is an integer, not an assumed positive quantity.

Theorem 3.1 (divisor-channel coordinates). *Set*

$$T_n = ne_{n-1} - e_n, \quad U_n = T_n - \sum_{\substack{d|n \\ 2 \leq d < n}} W_{d,n} U_d \quad (n \geq 2).$$

Then

$$M(U_n) = 0, \quad V_d(U_n) = (d! - 1)\mathbf{1}_{d=n}.$$

The vectors $e_1, U_2, U_3, \dots$ form an integral basis. Every finite vector has the unique finite expansion

$$\lambda = M(\lambda)e_1 + \sum_{d \geq 2} \frac{V_d(\lambda) - M(\lambda)}{d! - 1} U_d. \quad (5)$$

The observables, the unique expansion and the coefficient formula are **divisor channel coordinates**.

Proof. The moment of T_n is zero. The floor in $W_{d,n}$ changes between $n-1$ and n precisely when $d \mid n$, so

$$V_d(T_n) = (d! - 1)W_{d,n}\mathbf{1}_{d|n}.$$

The recursion cancels the proper divisor channels, proving the two identities by induction. The vectors $e_1, T_2, \dots, T_N$ form an integral triangular basis with final coefficients $1, -1, \dots, -1$. The change from T_n to U_n is integral triangular with unit diagonal. Hence $e_1, U_2, \dots, U_N$ is also an integral basis. Applying M and each V_d identifies its coefficients as in (5). For d beyond the support, $V_d = M$, so no infinite sum is required.

In particular, integrality of the displayed coefficients is a consequence of the basis calculation, not an extra divisibility assumption. $\square$

The prime translator is the case $U_p = T_p$ for a prime $p \geq 3$. The recursion also isolates composite channels; for example

$$U_9 = 9e_8 - e_9 - 5040e_2 + 1680e_3. \quad (6)$$

It has moment zero and only channel 9 is nonzero. Thus each channel can be adjusted separately by a multiple of its own denominator. The expansion is **composite translator nine** and the observable profile is **composite translator nine profile**.

3.1 The support restriction and the residual

Fix an integer $D \geq 2$. We first solve the channel equations in the auxiliary lattice and then impose the missing e_1 coordinate. These are two separate operations. Write

$$L_D = \text{lcm}_{2 \leq d \leq D}(d! - 1), \quad K_D = L_D e_1 - \sum_{d=2}^D \frac{L_D}{d! - 1} U_d.$$

Equation (5) implies both the channel congruence

$$V_d(\lambda) \equiv M(\lambda) \pmod{d! - 1} \quad (7)$$

in the equivalent integer-divisibility form [supported channel congruence](#), and the unique description of every low-channel kernel:

$$V_2 = \dots = V_D = 0 \iff \lambda = tK_D + \sum_{n>D} z_n U_n, \quad M = tL_D. \quad (8)$$

For the full residual

$$\mathcal{R}(\lambda) = \sum_{d \geq 2} \frac{V_d(\lambda)}{d! - 1},$$

we have $\mathcal{R}(e_1) = S$ and $\mathcal{R}(U_n) = 1$. Therefore

$$\mathcal{R}\left(tK_D + \sum_{n>D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n>D} z_n. \quad (9)$$

The series converges because $V_d = M$ beyond the support. This equation identifies the limit of channel adjustment: every zero-moment correction changes the residual by an integer. A further fractional cancellation coordinate does not arise from these corrections.

Only finitely many z_n are nonzero. Put $a_D = [e_1]K_D$ and $u_n = [e_1]U_n$. Admissibility on $n \geq 2$ becomes the single integer equation

$$ta_D + \sum_{n>D} z_n u_n = 0. \quad (10)$$

Consequently the attainable moments are

$$L_D \frac{g_D}{\gcd(g_D, a_D)} \mathbb{Z}, \quad g_D = \gcd\{u_n : n > D\}. \quad (11)$$

Indeed, finite integer combinations of the u_n form $g_D \mathbb{Z}$. The positive generator is attained because a gcd of any integer family is a finite integer combination of its members. An attaining vector is primitive: division by a common coefficient factor greater than one would preserve all zero channels and produce a smaller positive moment. Odd u_n vanish by induction. For a prime p , the sole nonzero proper-divisor contribution at $2p$ gives $u_{2p} = -2(2p)!/2^p \neq 0$. Every tail thus contains a nonzero coefficient, so $g_D > 0$.

A finite certificate for the infinite moment ideal. The scalar recurrence is

$$u_2 = 2, \quad u_n = - \sum_{\substack{d|n \\ 2 \leq d < n}} W_{d,n} u_d \quad (n > 2).$$

Once a divisor g controls the computed tail, only the terms with $d \leq D$ can supply a nonzero residue to a later coefficient. Those finitely many sources have a factorial cutoff:

$$k! \mid W_{d,dk} = \frac{(dk)!}{(d!)^k}.$$

The quotient by $k!$ counts partitions into k unordered blocks of size d .

Theorem 3.2 (finite determination of the moment ideal). *Choose a prime ℓ with $D/2 < \ell \leq D$ and put $H = D(2\ell - 1)$. Then*

$$g_D = \gcd(u_{D+1}, \dots, u_H), \quad H < 2D^2.$$

The statement is [finite channel moment certificate](#).

Proof. Since $D < 2\ell \leq D(2\ell - 1) = H$, the finite interval includes the nonzero term $u_{2\ell} = -(2\ell)!/2^{\ell-1}$. Thus its gcd g is positive and $g \mid (2\ell)!$. For $n > H$ and a divisor $d \leq D$, the integer n/d is at least 2ℓ . Hence $g \mid (n/d)! \mid W_{d,n}$. Every early source is divisible by g . Strong induction handles the remaining divisors $D < d < n$, whose coefficients are already divisible by g . Thus g divides the entire tail. Bertrand's postulate supplies ℓ for $D \geq 3$; take $\ell = 2$ for $D = 2$. Finally $H \leq D(2D - 1) < 2D^2$.

This is a quadratic upper horizon, not an assertion that the initial block through $2D$ always determines the tail gcd. $\square$

Theorem 3.3 (quotient-band breakpoint). *Let λ be a finitely supported integer vector, let $d \geq 2$ and $k \geq 0$ be integers, and suppose each index n in its support satisfies $kd \leq n < (k+1)d$. Then*

$$M(\lambda) = (d!)^k V_d(\lambda).$$

In particular, support in $[d, 2d)$ and $V_d(\lambda) = 0$ force $M(\lambda) = 0$. If all supported indices are at least d , $V_d(\lambda) = 0$ and $M(\lambda) \neq 0$, some supported index is at least $2d$.

Proof. On the stated band $\lfloor n/d \rfloor = k$, so $n! = (d!)^k W_{d,n}$. Sum against λ_n ; the final assertion follows by contraposition from the first-band case. $\square$

The exact factorisation is checked for every quotient band at [the band identity](#), and its zero-channel consequence is [band cancellation](#). The first-band form is explicit at [first-band factorisation](#); the final breakpoint alternative is [breakpoint witness](#).

The earlier envelope $\Lambda_n = \text{lcm}(1, \dots, n) \mid u_n$ remains useful for other certificates. Finite-source checks can end sooner because only the sum of the sources has to vanish modulo g . The even-coefficient sign pattern supplies a nonzero term in every tail. Together with (11) this determines the unrestricted minimum positive moment for every fixed depth. For example, $L_4 = 115$, $a_4 = -55$, $u_6 = -180$, $u_8 = -4200$ and $23u_6 - u_8 = 60 \mid \Lambda_9$ give minimum moment 1380, attained by $12K_4 + 253U_6 - 11U_8$.

On support through 6 the minimum is instead 4140. Minimum moment and minimum upper support are different objectives.

The first minimum allows arbitrary finite upper support; the second fixes an additional support cap. Neither optimisation can be substituted for the other in a residual estimate. Neither computation supplies a cofinal real-residual gap, and neither alters the two finite denominator exclusions.

The problem is not reduced to a finite search by the finite horizon: that horizon certifies the algebraic moment ideal at one fixed depth, whereas irrationality still requires an unbounded real-residual argument.

4 The moment cost and the choice of support

On the actual support $n \geq 2$, one has the stronger congruence $12 \mid M - 2V_2$. For $n = 2, 3$, $n! = 2W_{2,n}$. For $n \geq 4$, both $n!$ and $2W_{2,n}$ are divisible by 12: writing $n = 2k$ or $2k + 1$, $W_{2,2k} = k! \prod_{j=1}^k (2j - 1)$ is divisible by 6 for $k \geq 2$. Since every $d! - 1$ is coprime to 6, low-channel annihilation implies

$$12L_D \mid M. \quad (12)$$

The factor 12 is sharp at $D = 2$ and $D = 3$, witnessed respectively by $-6e_2 + e_4$ and $-6e_2 - 8e_3 + 5e_4$.

The factor L_D and the factor 12 combine because $\gcd(L_D, 12) = 1$; separate divisibility by two integers would not in general imply divisibility by their product.

4.1 Growth of the compulsory factor

The following elementary estimate quantifies this cost:

$$\liminf_{N \rightarrow \infty} \frac{\log L_N}{N^{3/2} \log N} \geq \frac{2\sqrt{2}}{3}. \quad (13)$$

The LCM asymptotic is proved here by an ordinary mathematical argument, and Lean checks it in `wcook04/plectis-erdos` as [common denominator growth liminf](#). The separate public Lean release `wcook04/plectis-erdos-lean` at commit `52f29ad1` states the bound (13), with the least common multiple of $n! - 1$ over $2 \leq n \leq N$ as the common denominator, as `common_denominator_growth` and in extended real form as `common_denominator_growth_liminf`. The statements are in the Comparator challenge file [PalomarCorpus/E68/Challenge.lean](#), where the proofs are left as sorry. The solution file [Solutions/PalomarCorpus/E68/CommonDenominatorGrowth.lean](#) closes both by transfer from `common_denominator_growth` in `ErdosProblems/Erdos68/PaperCompleteAsymptotics.lean` and `common_denominator_growth_liminf` in `ErdosProblems/Erdos68/PaperCompleteLiminf.lean` of the same release. That repository's Linux replay of Palomar's Comparator stage accepted the entry with the Lean kernel and with NanoDa ([run 34782407633](#)). The checked bound concerns a common denominator of the partial sums and bounds no reduced denominator of a rational representation of the series. The finite-block inequality has a separate Lean source, [finite terminal-block inequality](#). The divisibility $\gcd(i! - 1, j! - 1) \mid j!/i! - 1$ follows on multiplying $i! - 1$ by $j!/i!$ and subtracting $j! - 1$. Shared prime-power divisibility was already

used to control the spacing of factorial indices by Erdős and Stewart [13, §3, pp. 516–517]. The displayed subtraction is the case $P = -1$ of Luca and Shparlinski's polynomial-shift argument [9, proof of Lemma 5, p. 811], also used by Lai [8, proof of Lemma 2.4, display (2.5)]. These sources supply the inherited collision step, not the lcm asymptotic stated here. For positive integers x_i , the divisibility $\prod_i x_i \mid \text{lcm}(x_i) \prod_{i < j} \gcd(x_i, x_j)$ follows prime by prime: after ordering valuations, each except the maximum occurs at least once among the pairwise minima. On a terminal block of k terms, this inequality and $\gcd(i! - 1, j! - 1) \mid j!/i! - 1$ give

$$\log L_N \geq \sum_{n=N-k+1}^N \log(n! - 1) - \binom{k+1}{3} \log N.$$

Taking $k = \lfloor \alpha \sqrt{N} \rfloor$ gives leading coefficient $\alpha - \alpha^3/6$, maximised by $\alpha = \sqrt{2}$. For a fixed $P \in \mathbb{Z}[X] \setminus \{0\}$, the same ordinary argument gives this bound for $\text{lcm}_{n_0 \leq n \leq N} (n! + P(n))$. Here n_0 is chosen so that every factor exceeds 1 and every collision difference below is nonzero. Such a cutoff exists by [8, Lemma 2.1], where Lai adds the bound $n! + P(n) > 1$ to the nonvanishing statement of Lemma 3 of Luca and Shparlinski [9]. The lower cutoff matters: $P = -2$ makes the factor at $n = 2$ vanish. Multiplying the factor at i by $j!/i!$ and subtracting the factor at j gives

$$\gcd(i! + P(i), j! + P(j)) \mid P(j) - (j!/i!)P(i).$$

For $n_0 \leq i < j \leq N$ the right side is a nonzero integer of absolute value $O_P(N^{j-i+\deg P})$, so the block loses an additional $O_P(k^2 \log N)$, which is lower order. For $P = 0$, the lcm is only $N!$. The non-polynomial perturbation $n! + 2^n - 1$ requires a different elimination argument involving three hits and multiplicative order [14, Lemmas 2.1–2.3]; it is not covered by the polynomial-shift proof. These are common-denominator estimates. For $N \geq 3$, any common divisor of $(N-1)! - 1$ and $N! - 1$ divides $N-1$ by subtraction, and is coprime to $N-1$ because $N-1 \mid (N-1)!$. Thus the two denominators are coprime. Since $S - H_N > 1/((N+1)! - 1)$,

$$L_N(S - H_N) > \frac{((N-1)! - 1)(N! - 1)}{(N+1)! - 1} \rightarrow \infty.$$

Even two terms therefore obstruct clearing by the full common denominator; (13) gives the stronger quantitative cost. Spacing methods for factorial congruences [16, Lemma 2] and the fixed-value multiplicity bound [17, arXiv v1, Theorem 12, p. 16] are relevant comparisons, not ingredients of this proof.

4.2 Remote primitive kernels

Support location imposes a different requirement from low moment. Let $D, r \geq 2$ and let L be a positive multiple of $2, \dots, D$. Put

$$i_j = r + jL \quad (0 \leq j < D), \quad N = r + (D-1)L, \quad \alpha_d = (d!)^{L/d},$$

$$H(X) = \prod_{d=2}^D (\alpha_d X - 1) = \sum_{j=0}^{D-1} h_j X^j, \quad A = \prod_{d=2}^D \alpha_d.$$

The primitive kernel on this grid is

$$\lambda_j^* = \frac{N!h_j}{Ai_j!}, \quad M = N! \prod_{d=2}^D (1 - 1/\alpha_d) > 0.$$

Its final coefficient is 1.

Consequently its integer coefficients have gcd one. This explicit primitivity does not establish that it has the smallest possible support among all low-channel kernels. The channel equations are polynomial vanishing at the distinct points α_d^{-1} . Integrality follows by expanding h_j/A into products of reciprocal α_d : each product divides the factorial quotient $N!/i_j!$. The minimum step is $L = \text{lcm}(2, \dots, D)$.

The stronger divisibility

$$r! \prod_{d=2}^D (L/d)! \prod_{d=2}^D (\alpha_d - 1) \mid M$$

follows by counting partitions into the specified equal-sized blocks. In particular $\lfloor \sqrt{N}/2 \rfloor! \mid M$: if $v = \max(r, L/2)$, then $N < 4v^2$ and $v! \mid M$. Thus these moments absorb every fixed denominator as $N \rightarrow \infty$.

This assertion means that for each fixed $q \geq 1$ there is a threshold, depending on q , beyond which $q \mid M$. It is not a uniform estimate for denominators that grow with the support. On the factorial-grid Cramer vector the residual differs from a nonzero determinant times S by an integer: this is the [Cramer residual identity](#). Cofinal residual nonintegrality on this family is sufficient for irrationality; the construction does not establish that nonintegrality.

5 Cancellation in a reduced prefix

A common-denominator bound does not determine the denominator after addition. For linear forms in p -adic zeta values, with $p \geq 5$ a fixed prime and n the index of the form, Lai, Lupu and Sprang quantify a gap of this kind. They bound the denominators of the coefficients by powers of $\text{lcm}(1, \dots, n)$ and show that the bound may be divided by a product Φ_n of prime powers [10, §5]. The growth rate of Φ_n [10, Lemma 7.3] enters the inequality [10, (8.1)] under which the rescaled forms, which have integer coefficients, meet the irrationality criterion [10, Lemma 2.1, p. 3] that they quote from Lai [11, Lemma 2.1, p. 4]: nonzero values whose p -adic size, multiplied by the largest absolute value of a coefficient, tends to zero. For the prefix sums no asymptotic saving of this kind is established here; the first-layer test below decides, one prime at a time, whether the largest power of that prime survives reduction. This paragraph compares the two methods; no p -adic theorem is applied to S .

The test is the top valuation layer of the reciprocal-sum valuation formula of Louwsma and Martino [7, Lemma 4.1, p. 10], written out here for the denominators $n! - 1$. Put $L_M = \text{lcm}_{2 \leq n \leq M} (n! - 1)$ and $A_M = \sum_{n=2}^M L_M / (n! - 1)$. For a prime q with attained maximum $e = \max_{2 \leq n \leq M} v_q(n! - 1) > 0$, set

$$I_{q,e}(M) = \{n \in [2, M] : v_q(n! - 1) = e\}, \quad R_{q,e}(M) = \sum_{n \in I_{q,e}(M)} \left(\frac{n! - 1}{q^e} \right)^{-1} \pmod{q}.$$

After multiplication by L_M , all lower-valuation terms vanish modulo q . Hence

$$A_M \equiv (L_M/q^e)R_{q,e}(M) \pmod{q},$$

and L_M/q^e is a unit. Consequently

$$v_q(\text{den } H_M) = e \iff R_{q,e}(M) \not\equiv 0 \pmod{q}. \quad (14)$$

The equivalence, with the maximum and its attainment taken from the actual prefix lcm, is [maximal prime power survival](#).

The following factorial-gap examples come from an exact modular scan; they show why the weights matter. At $M = 138$, the maximal 139-hits are 69, 122, 137, with cofactor residues 6, 49, 73, and

$$6^{-1} + 49^{-1} + 73^{-1} \equiv 0 \pmod{139}.$$

At $M = 2592$, the maximal 2593-hits are 349, 2243, 2591, with residues 1508, 1566, 1678 and reciprocal sum zero. All these attained exponents are one, so the respective prime disappears completely from the reduced prefix. Singleton maximal support cannot cancel, whereas a hit-count bound alone cannot determine a weighted sum. Even survival must subsequently be coupled to a real strict-successor inequality.

6 One joint collision and residue estimate

The whole-prefix construction separates the common collision factor from prime-power layers with a unique maximal owner. For a natural parameter $p \geq 3$, write $d_n = n! - 1$ and put

$$\begin{aligned} F_p &= (p-1)!, \quad D_p = \text{lcm}_{2 \leq i < j \leq 2p-1} \gcd(d_i, d_j), \\ C_p &= \text{lcm}(F_p, D_p), \quad L_p^{\text{blk}} = \text{lcm}(F_p, d_2, \dots, d_{2p-1}), \\ R_p &= L_p^{\text{blk}}/C_p, \quad T_p = \sum_{n=2}^{2p-1} L_p^{\text{blk}}/d_n, \quad \rho_p = (-T_p) \bmod R_p. \end{aligned}$$

Every prime dividing R_p has a unique denominator attaining its maximal valuation. Reducing T_p modulo that prime leaves one unit term, giving $\gcd(T_p, R_p) = 1$. Thus $0 < \rho_p < R_p$ whenever $R_p > 1$.

Proposition 6.1 (global complementary-residue criterion). *If arbitrarily large natural parameters $p \geq 3$ satisfy*

$$R_p > 1, \quad (2p+1)L_p^{\text{blk}} < 2p^2(2p-1)!\rho_p, \quad (15)$$

then S is irrational.

The criterion at natural parameters is [global complementary criterion nat](#).

Proof. Suppose $S = a/q$. For large p , $q \mid F_p \mid C_p$, so $L_p^{\text{blk}}S$ is an integer multiple of R_p . Therefore

$$A = L_p^{\text{blk}}(S - H_{2p-1})$$

is a positive integer congruent to $-T_p$ modulo R_p , and $A \geq \rho_p$. The positive tail bound

$$S - H_{2p-1} < \frac{2p+1}{2p^2(2p-1)!}$$

and (15) give $A < \rho_p$, a contradiction. $\square$

Let $\tilde{C}_p = C_p/F_p$ and $\eta_p = \rho_p/R_p$. The inequality is exactly

$$\log \tilde{C}_p - \log \eta_p < \log \frac{2p^2(2p-1)!}{(2p+1)(p-1)!}. \quad (16)$$

The right side is $p \log p + (2 \log 2 - 1)p + O(\log p)$. The modulus R_p cancels: increasing private support alone does not improve this comparison. Coprimality permits $\rho_p = 1$, so the complementary gap needs its own quantitative control. Both losses must be controlled at the same parameters. The [unit-factor pair floor](#) splits that scale test into the complementary-residue inequality and a local collision-core cap. A joint mean bound below $(1 - \varepsilon)p \log p$ on the same nonempty sets in $[X, 2X]$ would suffice; two unrelated cofinal sets can be disjoint.

7 The remaining real comparison

The exact target remains the [strict-successor criterion](#):

$$(\forall B)(\exists m > B) \ m \nmid Z_m. \quad (17)$$

Two sufficient approaches now have explicit, different missing estimates. For the compressed primitive grids, the moments already absorb every fixed denominator. With positive moment M and upper support N , write

$$A_N = \sum_{d=D+1}^N \frac{V_d(\lambda)}{d! - 1}, \quad \mathcal{R}(\lambda) = A_N + M \sum_{d>N} \frac{1}{d! - 1}.$$

The sufficient strict comparison is

$$\frac{2M}{(N+1)! - 1} < \lfloor A_N \rfloor + 1 - A_N. \quad (18)$$

It puts the residual below the next integer and above A_N . Proving this on an unbounded grid family would suffice. The finite signed block A_N is essential. The second approach asks for the simultaneous collision and complementary-gap estimate (16). Neither estimate is established cofinally here.

In the first route $M > 0$ is essential for the stated one-sided tail comparison. The signed prefix A_N itself need not be positive. In the second route the core estimate and residue estimate must hold at the same parameters; separate unbounded families do not suffice.

The limit of short supports. Nonzero-moment kernels with $N = D + O(1)$ cannot meet (18). For $N = D + O(1)$, the compulsory divisibility $L_D \mid M$ and (13) give

$$\frac{2|M|}{(N+1)! - 1} \rightarrow \infty,$$

whereas the right side of (18) is at most one. A useful family must therefore allow a larger upper support and still control its finite signed block.

Large denominator valuations alone also leave the real gap uncontrolled: the model fractions $1/q^e$ tend to zero. The exact [zero-branch / lower-endpoint cylinder](#) and doubled-prime tests require joint numerator or predecessor information. Their thresholds, the [adjacent-window collapse](#), fixed-owner absorption, and the strongest counter-models are retained in the long reasoning record.

8 The finite evidence

An exact non-unit carry at the prime index 67 already forces [every rational denominator to be at least 67](#). The exact GMP carry certificate covers $3 \leq m \leq 300000$. Its unit carries occur at

$$52, 591, 1030, 1407, 1438, 2164, 4258, 10991, 21236.$$

In particular $b_{300000} \neq 1$, yielding $q \nmid 299999!$ by (1). This excludes every divisor of that factorial, including large divisors. It does not exclude all integers with small prime factors, since their multiplicities can be too large. The range through 300000 is a fresh exact GMP replay in this revision and lies outside the Lean development. The long record gives the receipt's digests and a separate replay through 4000. The certificate's event-trace, unit carries and terminal enclosure match the previously retained receipt; the published driver and backend hashes are the SPDX-bearing public copies.

The separate 80000-bit exact rational enclosure forces 23449 common continued-fraction partial quotients. The continued-fraction recurrences, determinant identity and fractional transformation [12, §1.12(ii), (1.12.5)–(1.12.7), (1.12.20)–(1.12.21)] turn this common prefix into the bound $q \geq 2^{39990} > 10^{12038}$; the enclosure, the prefix length and the exponent are outputs of the computation. The open-versus-closed complete-quotient endpoint convention matters for a sharper extraction; no stronger numerical floor is asserted here. The exact sources, enclosure and integer-computation receipts are retained with the long record. Neither finite calculation supplies (17).

A Guide to the formal sources

The mathematical proofs above are independent of the source-link audit. Links carrying the manuscript commit identify historical source locations; they do not assert that every linked module was compiled together. The accompanying revision bundle gives a declaration-level ledger for the supplied public snapshot 3d6d938d696f: source presence, recorded build inclusion, and computational evidence are separate fields.

The common-denominator liminf, moment horizon and moment-ideal modules have recorded checked-build entries in that ledger. Other supplied modules, including PaperCompleteExisting and the candidate finite-size certificate, remain outside

the recorded checked build. This revision replayed lake build ErdosProblems Erdos249257 on Lean 4.29.1 and mathlib 5e932f97dd25535344f80f9dd8da3aab83df0fe6; the library AxiomAudit prints the imported paper-linked declarations against the standard axioms propext, Classical.choice, and Quot.sound.

The continued-fraction enclosure, the small arithmetic witnesses, and the full carry census through 300000 have fresh independent integer replays. The earlier replay through 4000 remains a separate smaller certificate. The complete historical source catalogue is preserved in the handover rather than interleaved with the mathematical argument. Production PDF compilation with the repository preamble, fresh formal checking and a matching full carry receipt remain release gates.

Source-current statements and evidence boundaries. The finite divisor-event identities, isolated-channel recursion, vanishing moment, individual channel values, and the factor 12 are checked in [divisor channel basis](#). The unique integral basis expansion, attainable-moment formula and compressed primitive-grid construction are ordinary proofs, described in [divisor channel basis](#) and, for the grid construction, in [compressed primitive channel kernel](#). The quadratic stopping theorem is checked in [complete moment horizon](#), together with the scalar recurrence, the equal-block divisibility and the prime anchor it uses. The earlier lcm-envelope certificate and depth-4 moment 1380 are ordinary proofs with exact checks in [check moment saturation](#). The finite-source checks are in [check finite source](#). The inherited generic lemmas in [tail ideal certificate](#) assume the lcm envelope; they do not formalise the factorial arithmetic. The growth liminf is the ordinary asymptotic consequence of the finite terminal-block inequality cited above, and Lean checks it as [common denominator growth liminf](#). The checked counterpart of the growth liminf in the public Lean release wcook04/plectis-erdos-lean at commit 52f29ad1 is cited at (13). Formal proof checking does not assess external novelty or the significance of the remaining hypothesis. The complete source register, receipts and failed-route calculations belong to the long record.

Pinned declaration index. The public ErdosProblems.Erdos68 package contains the checked source for this note. The pinned snapshot contains fifteen cited modules: AdjacentUnitCarryWindow, CanonicalFactorialDigits, ChannelBreakpointRigidity, ChannelIntegralCongruence, DivisorFactorialCentre, EndpointWeightedPrivateSupport, FactorialCarry, FactorialChannelCertificate, FactorialGapPlateauCore (isolation pin; FactorialZeroPlateau is the default-root duplicate), FiniteDefectAutomaton, GapScalarNormalForm, PrimeThresholdParity, PrimeUnitTranslator, PrimeZeroBranch, and StrictSuccessorArithmetic. Only these public modules belong to the manuscript source surface; no private auxiliary digit-rigidity file is cited or projected. The public root imports every cited module. The declaration table below is pinned to the shared formal-source commit used throughout this problem-note series.

- [adjacent-window collapse](#)
- [two step den mul transition normalizers](#)
- [adjacent unit carry window den eq two step den](#)
- [adjacent unit carry window offset eq two step reduced numerator](#)
- [denominator at least the carry index](#)

- denominator at least 67
- strict-successor criterion
- cofinal non-unit carries imply irrationality
- zero-branch / lower-endpoint cylinder
- unit-factor pair floor
- Cramer residual identity
- residual centre term
- factorial coeff term
- residual centre
- factorial coeff rat
- factorial eq mul pred factorial
- pred div eq of not dvd
- pred div add one eq of dvd
- residual centre term of not dvd
- residual centre term of dvd
- residual centre term self
- factorial coeff term self
- residual centre recurrence
- weights are integers
- channel weight
- channel weight mul denominator
- channel numerator
- factorial moment
- channel event
- adjacent differences affect only divisor channels
- lambda34
- lambda34 apply three
- lambda34 apply four
- lambda34 channel two
- lambda34 factorial moment
- lambda34 channel three

- [lambda34 channel four](#)
- [lambda34 channel ge five](#)
- [lambda34 residual of tail](#)
- [lambda34 residual bounds](#)
- [square subsequence radius cubic lower](#)
- [no eventual square subsequence cubic upper](#)
- [not is little o square subsequence radius](#)
- [square subsequence radius three halves lower](#)
- [no eventual square subsequence three halves upper](#)

Source-current companion orbit and Comparator routes. The Comparator isolation pin for the carry equivalences is [factorial gap plateau core](#). [Factorial zero plateau](#) is the default-root duplicate of that family and is not the isolation source. The complete infinite rationality boundary is checked in [companion orbit rationality](#). Its paper-facing endpoints are [not irrational factorial gap series](#) iff eventually [companion floor \$-2\$](#) and [irrational factorial gap series](#) iff cofinal [companion floor](#) misses. The coherent Comparator composite [companion-orbit complete characterisation](#) routes to Theorem 2.1 and Remark 2. The moving-factor Comparator endpoints are recorded in the long reasoning record as the moving-factor scale split, the split-factor normalised collision, and the fixed-owner absorption boundary. The separate public Lean release `wcook04/plectis-erdos-lean` at commit `52f29ad1` selects those three endpoints as `movingPrivateFactorScaleSplit_implies_irrational`, `splitFactorNormalizedCollision_implies_irrational` and `fixedOwnerPair_eventually_absorbed` in `PalomarCorpus/E68/Challenge.lean`. These source-current modules, Comparator packages, and this manuscript stage require one common immutable public checkpoint before terminal external replay or Palomar readiness is claimed; for the eight-entry release that checkpoint is commit `52f29ad1`, whose Linux replay of Palomar's Comparator stage accepted the E68 entry (run `34782407633`, cited at (13)). The unconditional correction (4) is checked in [companion constant carry law](#), and the pole-residue identity and the two displayed reciprocal equalities are checked in [prime pole criterion](#). The complete maximal-hit data come from the separate exact modular scan; the carry census and continued-fraction certificate use exact integer computation outside Lean. The inherited public source pin is `99f4bf47422a`; source-current additions and that snapshot require one immutable release before the links represent a replay of the whole note.

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Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.